Skills Summary

Planning Constructions and Justifying Them

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing a Construction Plan

Legal moves: set the compass to a span; strike an arc from a marked point; draw a line, ray or segment through two marked points; mark a crossing. Measuring with a ruler or protractor is not a move.

A plan says three things per step: where the compass point sits, what the span is, and whether the span changes. "Keep the same setting" is information, not decoration — it is what the justification will lean on.

Example: Plan a copy of $\overline{AB}$, where A is at $(0,0)$ and B is at $(8,6)$, onto a ray starting at C.
Step 1. The copy has to be congruent, not approximately equal, so the span itself must be transferred — that rules out reading a length and re-drawing it.
Step 2. Draw the ray from C; open the compass from A to B and do not change it; put the compass point on C and strike an arc across the ray; mark the crossing. Its distance from C is $\sqrt{64+36}$.
$\Rightarrow$ the copy must measure **10**

Try it: You copy $\overline{PQ}$ with P at $(1,1)$ and Q at $(4,5)$. What length must the copy have?

check: **5**

Skill 2 — Justifying with Equal Radii

The whole argument in one line: arcs of equal radius create congruent segments; congruent segments create congruent triangles; congruent triangles hand you the angle or the perpendicular you claimed.

Template: name the equal radii, name the shared side, cite side-side-side, then name the corresponding part you wanted. A justification that only re-tells the drawing has skipped every one of those.

Example: Arcs of one radius centred at $A(0,0)$ and at $B(8,0)$ cross at $P(4,3)$. Justify that P is on the perpendicular bisector of $\overline{AB}$.
Step 1. The two arcs were struck with one compass setting, so the claim to argue from is that P is the same distance from each endpoint — not that the picture looks symmetric.
Step 2. From A the run is 4 and the rise is 3; from B the run is 4 the other way and the rise is 3. Both give $\sqrt{16+9}$, so the two distances are equal and P is equidistant from the endpoints.
$\Rightarrow$ $PA = PB = \mathbf{5}$, so P lies on the perpendicular bisector.

Try it: Arcs of one radius centred at $(0,0)$ and at $(10,0)$ cross at $(5,12)$. Find the distance to each endpoint.

check: both are **13**

Skill 3 — Checking a Construction on Coordinates

Distance formula: the run and the rise are the legs of a right triangle, so the distance between two points is $\sqrt{(\text{run})^2 + (\text{rise})^2}$.

Use it to audit a claim: a midpoint claim means two equal halves; an equidistant claim means two equal distances; a dilation claim means the two distances from the centre are in the ratio of the scale factor.

Example: A construction claims $N(3,5)$ is the midpoint of $\overline{AB}$, where A is at $(-1,2)$ and B is at $(7,8)$. Check it.
Step 1. A midpoint claim is two claims at once — equal halves and lying on the segment — so compare the two halves rather than trusting the picture.
Step 2. From A to N the run is 4 and the rise is 3; from N to B the run is 4 and the rise is 3 again. Both halves are $\sqrt{16+9}$, and the matching run and rise also show N is on the segment.
$\Rightarrow \quad AN = NB = \mathbf{5}$

Try it: Is $(3,5)$ the midpoint of the segment from $(-3,1)$ to $(9,9)$? Give each half to two decimal places.

check: both halves are **7.21**

(!) Watch out: equal distances alone prove a point is on the perpendicular bisector, but they do *not* prove a point is a midpoint — for that the point must also lie on the segment. And a plan is not finished until it says which spans stay unchanged.